HydroPlan-CA

Technical Documentation

Version 1

Developed as part of the project **Hydro4U**

funded under the Horizon 2020 research and innovation programme under grant agreement No. 101022905

May 2026

Disclaimer

HydroPlan-CA is intended as a decision support and exploratory tool only. The information and results presented are based on the best available data at the time of development, but data quality, resolution, and completeness may vary across regions. The tool does not replace detailed site-specific studies, technical assessments or environmental impact assessments. The Hydro4U project partners assume no liability for decisions made based on the use of this tool or the data displayed within it. Users are encouraged to verify all information with original data providers and relevant authorities before applying it in planning or decision-making contexts. The European Commission is not responsible for the content of this tool or for the use or interpretation of the information presented therein.

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Contents

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1 Introduction

HydroPlan-CA is a web-based Decision Support Tool designed to support evidence-based planning for sustainable hydropower development in Central Asia. The tool has been developed within the EU-funded Horizon 2020 project *Hydro4U*, with the aim of improving access to information relevant for hydropower planning in the region.

In many parts of Central Asia, the limited availability of reliable and harmonized data hampers informed decision-making in the hydropower sector. ***HydroPlan-CA*** addresses this challenge by integrating hydrological, technical, and environmental information into an interactive online Geographic Information System (GIS). By bringing together diverse datasets within a single platform, the tool is primarily intended as a first screening tool for hydropower-relevant decision support. The data provided can also be used for other assessments related to freshwater resources.

This document is intended to support users in understanding the origin, structure, and meaning of the data underlying the application, as well as the functionalities of the tool.

HydroPlan-CA can be accessed using the following link:

https://experience.arcgis.com/experience/d7f8330d2ed5471d8bed8d

The structure of the **Technical Documentation** is as follows:

- **Chapter 2** provides an overview of the datasets included in the ***Base Data*** *section* of ***HydroPlan-CA***, along with their descriptions and references where external data sources are used. It also describes all **attributes** associated with the ***River Network*** dataset.
- **Chapter 3** provides an overview of the **tool's interface and functionalities**.

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2 Data description

2.1 Base data description & references

This section provides an overview of the ***Base Data***, including references to the original data sources. The data can be accessed through the contents panel (see Section 3.4) and is organized into the following categories:

- **Hydrology**
- **Landcover**
- **Ecology**
- **Dams and Hydropower Plants**
- **Geology and Geohazards**

Most of the data cannot be downloaded directly from the online tool; however, all datasets are available free of charge from the original sources listed.

Note: *For most* ***Base Data*** *layers,* ***additional information*** *can be accessed by* ***clicking on the respective spatial data*** *within the map.*

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Hydrology	**Major Basins**	**Description:** Spatial delineation of major river basins in Central Asia. **Sources:** Compiled from multiple sources.
	Gauges	**Description:** Locations of hydrological gauging stations across Central Asia, including long-term mean discharge values. Where available, time series data can be downloaded as .csv files when clicking on the ***Data Access*** link in the attribute table. The dataset was compiled from third-party sources. **Sources:** Marti, B., Siegfried, T., Yakovlev, A., Zhumabaev, A., Karger, D. N., Wakil, A. W., & Ragettli, S. (2023). *CA-discharge data set, scripts and raw data* (Version 1.0) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.7743778 Marti, B., Yakovlev, A., Karger, D. N., Ragettli, S., Zhumabaev, A., Wakil, A. W., & Siegfried, T. (2023). CA-discharge: Geo-located discharge time series for mountainous rivers in Central Asia. *Scientific Data*, *10*, 579. https://doi.org/10.1038/s41597-023-02474-8
Landcover	**Glaciated Areas**	**Description:** Inventory of glacier outlines representing the spatial extent of glaciers. **Source:** RGI 7.0 Consortium. (2023). *Randolph Glacier Inventory - A dataset of global glacier outlines* (Version 7) [Data set]. NSIDC. https://doi.org/10.5067/F6JMOVY5NAVZ
	Land Cover (Sentinel 2)	**Description:** Land cover dataset derived from multispectral satellite imagery, providing high-resolution (10 m) information on surface characteristics. The dataset classifies land surface types such as vegetation, water bodies, urban areas, and bare soil. **Source:** Esri & Impact Observatory. (2025). *Sentinel-2 10 m land use/land cover time series* (Version 003) [Data set]. https://www.arcgis.com/home/item.html?id=cfcb7609de5f478eb7666240902d4d3d

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Ecology	**Endangered Freshwater Species**	**Description:** The International Union for Conservation of Nature (IUCN) Red List dataset provides information on the conservation status of species worldwide. It classifies species into different threat categories based on criteria including population trends, habitat range, and threats. The map shows the geographic distribution ranges of freshwater species categorized as *Critically Endangered*, *Endangered* and *Vulnerable*, depicting the highest endangered category per area. **Source:** IUCN (2025). *The IUCN Red List of threatened species* [Data set]. Retrieved January 2025. https://www.iucnredlist.org
	IUCN **Protected Areas**	**Description:** The World Database on Protected Areas (WDPA) is the most comprehensive global dataset on protected areas, including terrestrial, inland waters, coastal and marine protected areas. It provides spatial and attribute data on the location, size, IUCN management category, status, designation year, and management of protected areas, supporting conservation planning, biodiversity monitoring, and policy development worldwide. The IUCN Protected Areas layer includes only those areas assigned to an official IUCN management category which were then grouped as follows: Group 1 (Ia, Ib, II), Group 2 (III, IV), and Group 3 (V, VI). **Source:** UNEP-WCMC & IUCN (2025). *Protected planet: the world database on protected areas (WDPA) and world database on other effective area-based conservation measures (WD-OECM)* [Data set]. Retrieved August 2025. https://www.protectedplanet.net/
	Ramsar Wetland Sites	**Description:** The Ramsar Site Information Service provides information on wetlands designated under the Ramsar Convention, an international treaty for the conservation and sustainable use of wetlands. The dataset includes details such as site boundaries, geographic coordinates, ecological characteristics, and conservation measures. **Source:** Ramsar Site Information Service (RSIS) (2024). *Ramsar Sites* [Data set]. Retrieved August 2025. https://rsis.ramsar.org

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Freshwater Ecoregions	**Description:** The Freshwater Ecoregions of the World (FEOW) dataset provides a global classification of freshwater ecosystems based on biogeographic patterns. It includes spatial data on ecoregions, species distributions, and ecological characteristics. This data can be used to support freshwater biodiversity conservation, research, and management efforts. **Source:** The Nature Conservancy and World Wildlife Fund (2013). *Freshwater ecoregions of the world* [Data set]. Retrieved May 2025. https://www.feow.org/
Freshwater Key Biodiversity Areas	**Description:** The Key Biodiversity Areas (KBA) dataset identifies sites critical for global biodiversity conservation. It includes spatial and ecological data on areas that meet specific criteria for species and habitat importance, supporting conservation planning, policy development, and sustainable management efforts worldwide. The map only includes areas related to freshwater ecosystems. **Source:** BirdLife International (2024). *World Database of Key Biodiversity Areas* [Data set]. Retrieved June 2024. https://www.keybiodiversityareas.org Developed by the KBA Partnership: BirdLife International, International Union for the Conservation of Nature, American Bird Conservancy, Amphibian Survival Alliance, Conservation International, Critical Ecosystem Partnership Fund, Global Environment Facility, Re:Wild (formerly Global Wildlife Conservation), NatureServe, Rainforest Trust, Royal Society for the Protection of Birds, Wildlife Conservation Society and the World Wildlife Fund.
World Heritage Sites (WHS)	**Description:** The WDPA dataset was filtered for areas designated as natural or mixed World Heritage Sites (WHS). **Source:** UNEP-WCMC, IUCN (2025). *Protected planet: the world database on protected areas (WDPA) and world database on other effective area-based conservation measures (WD-OECM)* [Data set]. Retrieved August 2025]. https://www.protectedplanet.net/

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Other Protected Areas	**Description:** WDPA entries with no IUCN category (i.e., *not assigned*, *applicable*, or *reported*) and *not designated* as Ramsar or WHS were classified as *Other protected areas*. Thereby, important national protected areas such as nature sanctuaries (e.g., *Zakaznik*) or strictly protected areas (i.e., *Zapovednik*) can also be mapped. **Source:** UNEP-WCMC, IUCN (2025). *Protected planet: the world database on protected areas (WDPA) and world database on other effective area-based conservation measures (WD-OECM)* [Data set]. Retrieved August 2025. https://www.protectedplanet.net/
River-Specific Ecological Information	**Description:** The *River Specific Ecological Information* summarizes the above-described ecological criteria for each respective river section. In addition, it shows whether the river segment is considered *free-flowing*, *rare*, or a *linking* section. Free-flowing rivers are defined as rivers sections with a long-term mean discharge > 1,000 m^3/s and a Connectivity Status Index (CSI) > 99. It also includes their still connected and intact tributaries (i.e., CSI = 100) based on Grill et al. (2019). Rare river sections are identified by combining information on slope-based fish regions following Huet (1949) and freshwater ecoregions as described by De Keyser et al. (2026). Linking sections are defined as those sections which link the aforementioned river types, and have a length of ≤10km. **Sources:** Hydro4U project De Keyser, J., Seliger, C., Hayes, D. S., Schwedhelm, H., Fuentes, P. O., Carey, J. I., Siegfried, T., Marti, B., & Habersack, H. (2026). A regional-scale framework for assessing sustainable hydropower potential based on open-access data: The case of Central Asia. *Applied Energy*, *414*, 127843. https://doi.org/10.1016/j.apenergy.2026.127843

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Dams and Hydropower Plants	**Hydro4U Demo and Planning Sites**	**Description:** The Hydro4U project demonstrated two innovative hydropower technologies (*Demonstration Sites*). In additional, the project replicated the roll-out of these technologies by developing detailed planning studies at three *Planning Sites*. **Source:** Hydro4U project
	HPPs Not Controlled by Storage Operation	**Description:** Hydropower plants which are not directly influenced by upstream storage operations. The dataset is derived from a compiled geodatabase of hydropower plants in Central Asia. **Sources:** De Keyser, J., Fuentes, P. O., Hayes, D. S., & Habersack, H. (2026). A review of hydropower in Central Asia: Past, present, and future. *Renewable and Sustainable Energy Reviews*, *226*, 116239. https://doi.org/10.1016/j.rser.2025.116239 De Keyser, J., Osuna Fuentes, P., Hayes, D. S., & Habersack, H. (2025). *HP:CA - A geodatabase on hydropower plants in Central Asia* (Version 1) [Dataset]. Zenodo. https://doi.org/10.5281/zenodo.13320645
	HPPs Controlled by Storage Operation	**Description:** Hydropower plants where electricity generation is directly influenced by upstream reservoir operations. The dataset is derived from a compiled geodatabase of hydropower plants in Central Asia. **Sources:** De Keyser, J., Fuentes, P. O., Hayes, D. S., & Habersack, H. (2026). A review of hydropower in Central Asia: Past, present, and future. *Renewable and Sustainable Energy Reviews*, *226*, 116239. https://doi.org/10.1016/j.rser.2025.116239 De Keyser, J., Osuna Fuentes, P., Hayes, D. S., & Habersack, H. (2025). *HP:CA - A geodatabase on hydropower plants in Central Asia* (Version 1) [Dataset]. Zenodo. https://doi.org/10.5281/zenodo.13320645

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Planned HPPs	**Description:** This dataset contains locations of planned hydropower plants identified during data compilation from various sources. The dataset is not exhaustive, and the planning stage varies significantly between projects. **Sources:** De Keyser, J., Fuentes, P. O., Hayes, D. S., & Habersack, H. (2026). A review of hydropower in Central Asia: Past, present, and future. *Renewable and Sustainable Energy Reviews*, *226*, 116239. https://doi.org/10.1016/j.rser.2025.116239 De Keyser, J., Osuna Fuentes, P., Hayes, D. S., & Habersack, H. (2025). *HP:CA - A geodatabase on hydropower plants in Central Asia* (Version 1) [Dataset]. Zenodo. https://doi.org/10.5281/zenodo.13320645
Dams	**Description:** This dataset is a compiled collection of dam locations across Central Asia, derived from multiple sources. Where information is available, the primary purpose of each dam is included as an attribute. In some cases, dams may have a secondary purpose (e.g., main purpose - irrigation, secondary purpose - energy generation). **Sources:** Zhang, A. T., & Gu, V. X. (2023). *Global Dam Tracker: A database of more than 35,000 dams with location, catchment, and attribute information* (Version v1) [Dataset]. Zenodo. https://doi.org/10.5281/zenodo.7616852 Lehner, B., ReidyLiermann, C., Revenga, C., Vorosmarty, C., Fekete, B., Crouzet, P., Doll, P., Endejan, M., Frenken, K., Magome, J., Nilsson, C., Robertson, J. C., Rodel, R., & Sindorf, N. (2011). *Global Reservoir and Dam Database, Version 1 (GRanDv1): Dams, Revision 01* (Version 1.01) [Dataset]. Palisades, NY: NASA Socioeconomic Data and Applications Center (SEDAC). https://doi.org/10.7927/H4N877QK J. Wang, B.A. Walter, F. Yao, C. Song, M. Ding, A.S. Maroof, J. Zhu, C. Fan, J.M. McAlister, S. Sikder, Y. Sheng, G.H. Allen, J.-F. Crétaux, Y. Wada, *GeoDAR: georeferenced global dams and reservoirs dataset for bridging attributes and geolocations* [Data set], (2022). https://doi.org/10.5194/essd-14-1869-2022 Scientific-Information Center ICWC. (n.d.) *Portal of Knowledge for Water and Environmental Issues in Central Asia (Reservoirs).* http://www.cawater-info.net/bk/1-1-1-1-3_e.htm

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Geology and Geohazards	**Geology**	**Description:** Global Lithological Map (GLiM), representing the distribution of surface rock types at ~1 km resolution. It classifies lithology into standardized categories such as *sedimentary*, *igneous*, *metamorphic rocks*, and *unconsolidated sediments*. **Source:** Hartmann, J., & Moosdorf, N. (2012). The new global lithological map database GLiM: A representation of rock properties at the Earth's surface. *Geochemistry, Geophysics, Geosystems, 13*(12). Dataset online available at: https://www.geo.uni-hamburg.de/en/geologie/forschung/aquatische-geochemie/glim.html
	Landslide Susceptibility	**Description:** Dataset derived using machine-learning techniques to map landslide susceptibility. It represents the likelihood of landslide occurrence based on environmental conditions such as slope, lithology, and rainfall. Values range from *null* to *very high* susceptibility and represent spatial likelihood rather than temporal probability. **Sources:** The World Bank. (2023). *Central Asia Landslide Susceptibility.* https://datacatalog.worldbank.org/search/dataset/0064119 Rosi, A., Frodella, W., Nocentini, N., Caleca, F., Havenith, H. B., Strom, A., Saidov, M., Bimurzaev, G. A., & Tofani, V. (2023). Comprehensive landslide susceptibility map of Central Asia. *Natural Hazards and Earth System Sciences*, 23(6), 2229–2250. https://doi.org/10.5194/nhess-23-2229-2023

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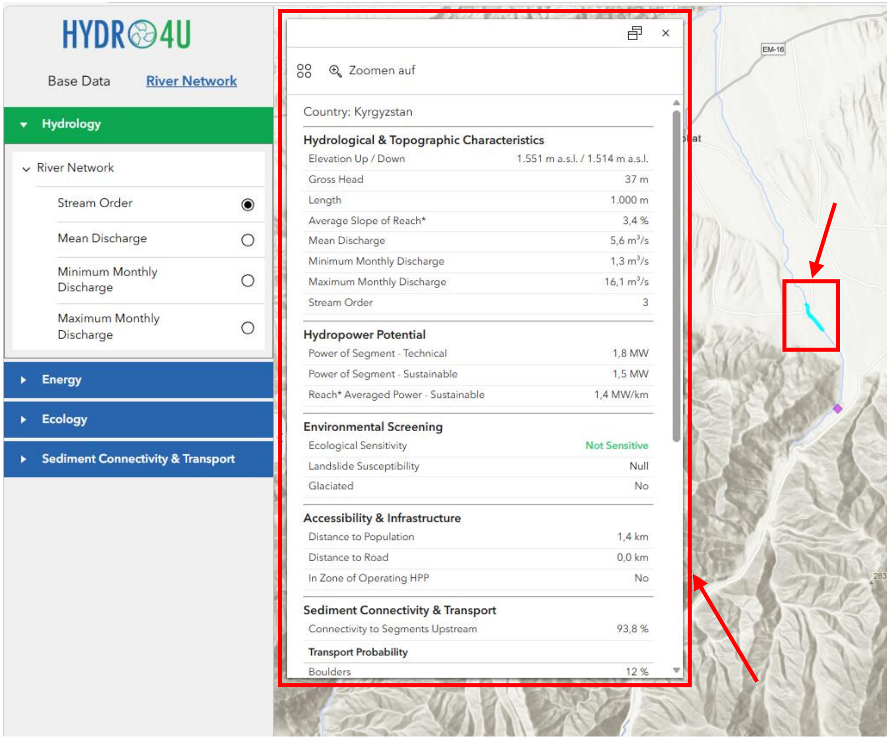
2.2 Attribute description of River Network data

The ***River Network*** data forms the **core component of the *HydroPlan-CA* tool**. To effectively use the tool and its interactive features, it is essential to correctly interpret the information provided within the attribute table which becomes available when clicking on a river segment within the application, and all attributes refer to the selected river segment.

How the ***River Network*** can be displayed based on selected attributes is described in Section 3.5. This section here provides the detailed description of all data fields.



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	Attribute (Unit)	Description
	Country	Indicates the country in which the river segment is located.
Hydrological & Topographic Characteristics	Elevation Up / Down (m a.s.l)	Elevation at the up-/downstream point of the river segment, derived from the hydrologically-conditioned Digital Elevation Model (DEM) from HydroSHEDS (Lehner et al., 2008).
	Gross Head (m)	Elevation difference between upstream and downstream points.
	Length (m)	Length of the river segment measured in the x-y plane.
	Average Slope of Reach (%)	Average slope of the river reach. ***Note:*** *A reach is defined as the sum of all connected river segments between two confluences.*
	Mean Discharge (m^3/s)	Mean annual discharge based on the Flo1K dataset (Barbarossa et al., 2018).
	Minimum / Maximum Monthly Discharge (m^3/s)	Long-term average of the minimum/maximum monthly discharge based on the Flo1K dataset (Barbarossa et al., 2018).
	Stream Order	Hierarchical classification of streams according to Strahler (Strahler, 1957). Headwater streams are classified as first order. When two streams of the same order meet, the stream order increases by one; when streams of different orders meet, the higher order is retained.
Hydropower Potential	Power of Segment – Technical (MW)	Technical hydropower capacity of the river segment based on hydraulic conditions, assuming a design discharge equal to Q30 (flow exceeded 30% of the time based on monthly discharge data).
	Power of Segment – Sustainable (MW)	Sustainable hydropower capacity of the river considering environmental flow release, assuming a design discharge equal to Q30 (flow exceeded 30% of the time based on monthly discharge data).
	Reach-Averaged Power – Sustainable (MW/km)	Sustainable hydropower potential normalized by reach length (expressed in km). ***Note:*** *A reach is defined as the sum of all connected river segments between two confluences.*

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Environmental Screening	Ecological Sensitivity	Ecological sensitivity classification (e.g., *not sensitive*, *sensitive*, *highly sensitive*, and *no-go*).
	Landslide Susceptibility	Indicates the level of attention required due to landslide risk, based on a landslide susceptibility analysis using the data provided in the ***Base Data***.
	Glaciated	Indicates whether the river segment intersects glaciated areas provided in the ***Base Data***.
Accessibility & Infrastructure	Distance to Population (km)	Distance to the nearest population center, based on OpenStreetMap data (OpenStreetMap contributors, 2025). The OSM population center data is limited to the following types (*fclass*): *national capital, city, town,* and *village.*
	Distance to Road (km)	Distance to the nearest road, based on OpenStreetMap data (OpenStreetMap contributors, 2025). The OSM road network is limited to the following types (*fclass*): *motorway, motorway_link, primary, primary_link, residential, secondary, secondary_link, service, tertiary, tertiary_link, track_grade1–3, trunk*, and *trunk_link.*
	In Zone of Operating HPP	Indicates whether the river segment lies within the area of an existing hydropower plant which were mapped based on the HP:CA dataset (includes. e.g., reservoir areas for storage plants or residual flow reaches for diversion plants) (De Keyser et al., 2025).
Sediment Connectivity & Transport	Connectivity to Segments Upstream (%)	Indicates the proportion of upstream river segments that are morphologically connected to a given segment and are therefore able to deliver sediment to it, relative to the total number of upstream river segments. Information is only available for four main Central Asian catchments: Amu Darya, Syr Darya, Chu-Talas, and Issyk-Kul.
	Transport Probability (%)	Probability that a given sediment size class is mobilized under channel-forming flow conditions (Q5; flow exceeded 5% of the time based on monthly discharge data). Sediment size classes include boulder (>200 to ≤630 mm), cobble (>63 to ≤200 mm), gravel (>2.0 to ≤63 mm), sand (>0.063 to ≤2.0 mm), and silt (>0.002 to ≤0.063 mm). The presence of individual grain-size classes at a given river segment is not considered. Information is only available for four main Central Asian catchments: Amu Darya, Syr Darya, Chu-Talas, and Issyk-Kul.
	Entrainment Probability (%)	Probability that a given sediment size class is eroded under channel-forming flow conditions (Q5; flow exceeded 5% of the time based on monthly discharge

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		data). Sediment size classes include boulder (>200 to ≤630 mm), cobble (>63 to ≤200 mm), gravel (>2.0 to ≤63 mm), sand (>0.063 to ≤2.0 mm), and silt (>0.002 to ≤0.063 mm). The presence of individual grain-size classes at a given river segment is not considered. Information is only available for four main Central Asian catchments: Amu Darya, Syr Darya, Chu-Talas, and Issyk-Kul.
	Deposition Probability (%)	Probability that a given sediment size class is deposited under channel-forming flow conditions (Q5; flow exceeded 5% of the time based on monthly discharge data). Sediment size classes include *boulder* (>200 to ≤630 mm), *cobble* (>63 to ≤200 mm), *gravel* (>2.0 to ≤63 mm), *sand* (>0.063 to ≤2.0 mm), and *silt* (>0.002 to ≤0.063 mm). The presence of individual grain-size classes at a given river segment is not considered. Information is only available for four main Central Asian catchments: Amu Darya, Syr Darya, Chu-Talas, and Issyk-Kul.

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3 Tool Functions

3.1 Map Navigation

To **change the map scale**, use the mouse scroll wheel to zoom in or out.

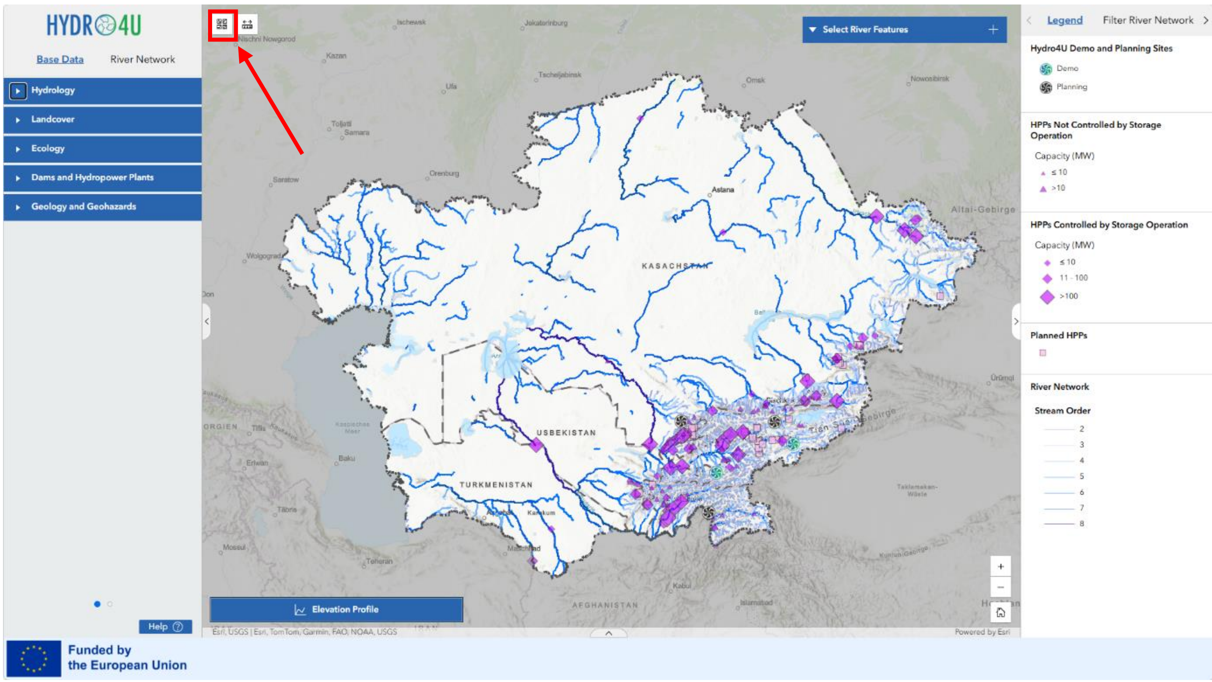
To **move across the map**, click and hold the left mouse button on the map. Drag the map in the desired direction and release the mouse button to stop moving.

To **return to the initial map extent**, click the ***Home button*** in the map toolbar (bottom right). The map will automatically return to its default starting view.

3.2 Change Basemap

The ***Basemap Gallery Widget*** is located in the **upper left corner of the map**. It allows users to change the background map displayed beneath the ***River Network*** data.

Click the widget to open a gallery of available basemaps. Click on a basemap thumbnail to apply it to the map. The selected basemap will immediately replace the current background map.



3.3 Measure Distances

The ***Measurement Widget*** is located in the **upper left corner** of the map (next to the Basemap Gallery widget) and allows users to measure distances and areas directly in the web map. Click the widget icon to open it and choose between two tools:

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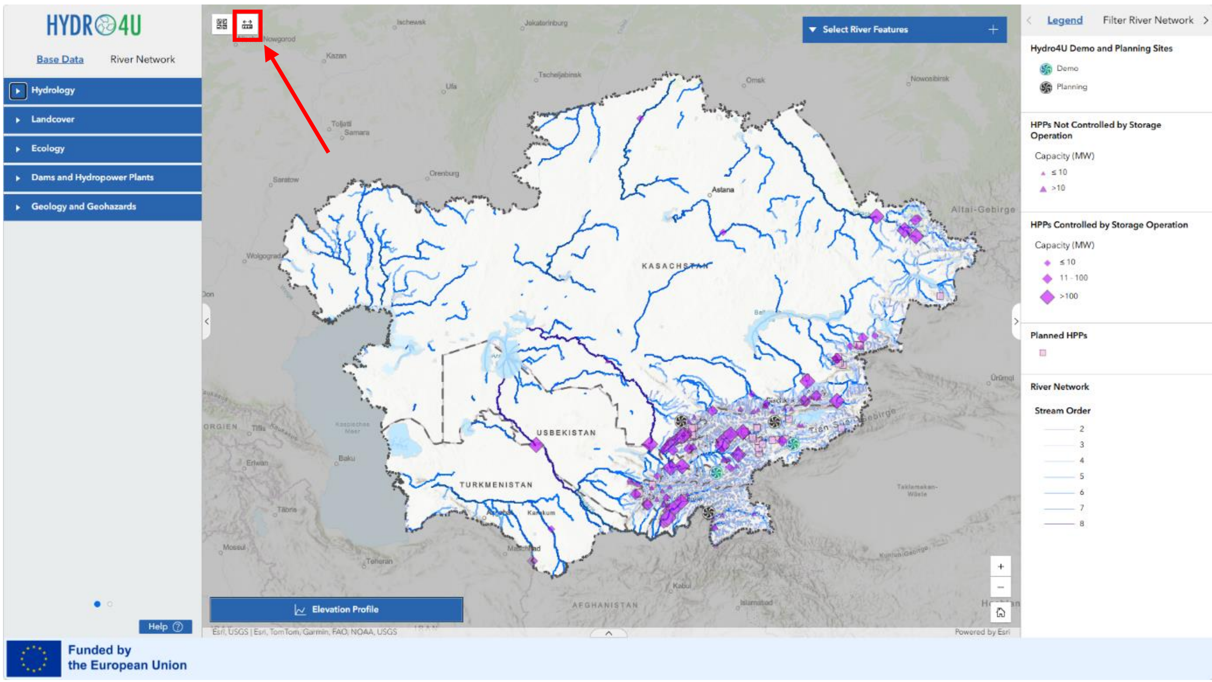
- Distance: Measure a linear distance. Click the map where you want to start the measurement and click once for every vertex of the line you want to measure. Double-click to complete the line.
- Area: Measure an area and its perimeter. Click the map where you want to start the measurement and click once for every vertex of the area you want to measure. Double-click to complete the area.

Users can click the **Unit** drop-down menu and select a different unit of measure. Measurement results convert to the newly selected unit.

Note: *The Measurement widget uses snapping. The pointer snaps to features on the map. You can press and hold the Ctrl key to temporarily turn off snapping.*



3.4 Base Data

The ***Base Data*** tab is located on **top of the left panel**. It provides access to a range of thematic background layers that give spatial information displayed on the map.

The layers within the ***Base Data*** tab are organized into five thematic categories:

- **Hydrology**
- **Landcover**
- **Ecology**
- **Dams and Hydropower Plants**
- **Geology and Geohazards**

Each category can be expanded or collapsed by clicking on the category header. This allows you to keep the panel tidy and only view the layers relevant to your current task.

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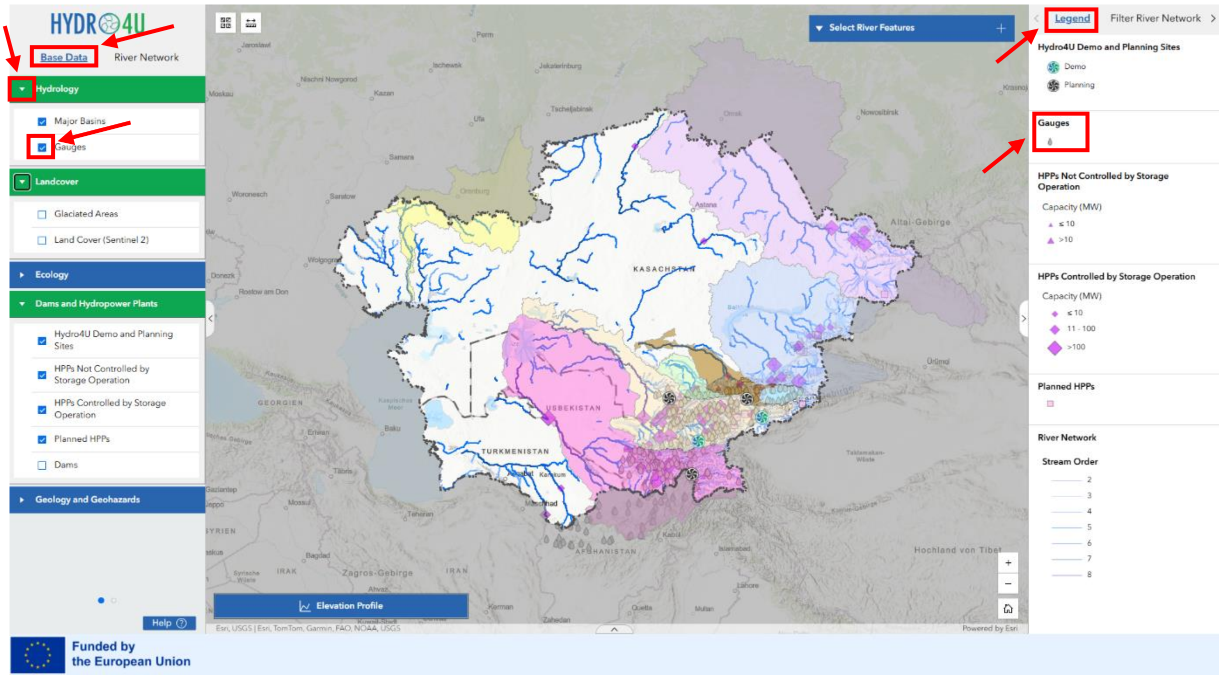
A detailed **description of all layers**, including references to the original data sources, can be found in **Section 2.1**.

To turn on or off a layer, click the **tick box** next to the layer name. A ticked box means the layer is currently visible on the map.

The legend for the active layers can be viewed in the **panel on the right side of the map**.



3.5 River Network Data

The ***River Network*** *tab* is located on **top of the left panel** next to the Base Data tab. It displays the ***River Network*** with information, organized into four thematic groups:

- **Hydrology**
- **Energy**
- **Ecology**
- **Sediment Connectivity & Transport**

Each group can be expanded or collapsed by clicking on the group header. Within each group, layers are further organized into sub-groups. To make a layer visible on the map, the individual layer must be activated using the bottom next to a ***River Network*** layer.

When clicking on the river network, a river segment of approximately 1 km in length is selected, and additional information for this specific segment is displayed in a pop-up box in the top left of the map. For a detailed description of each layer and its attributes, refer to **Section 0**. Each group represents the same underlying river network but visualizes different thematic attributes and information.

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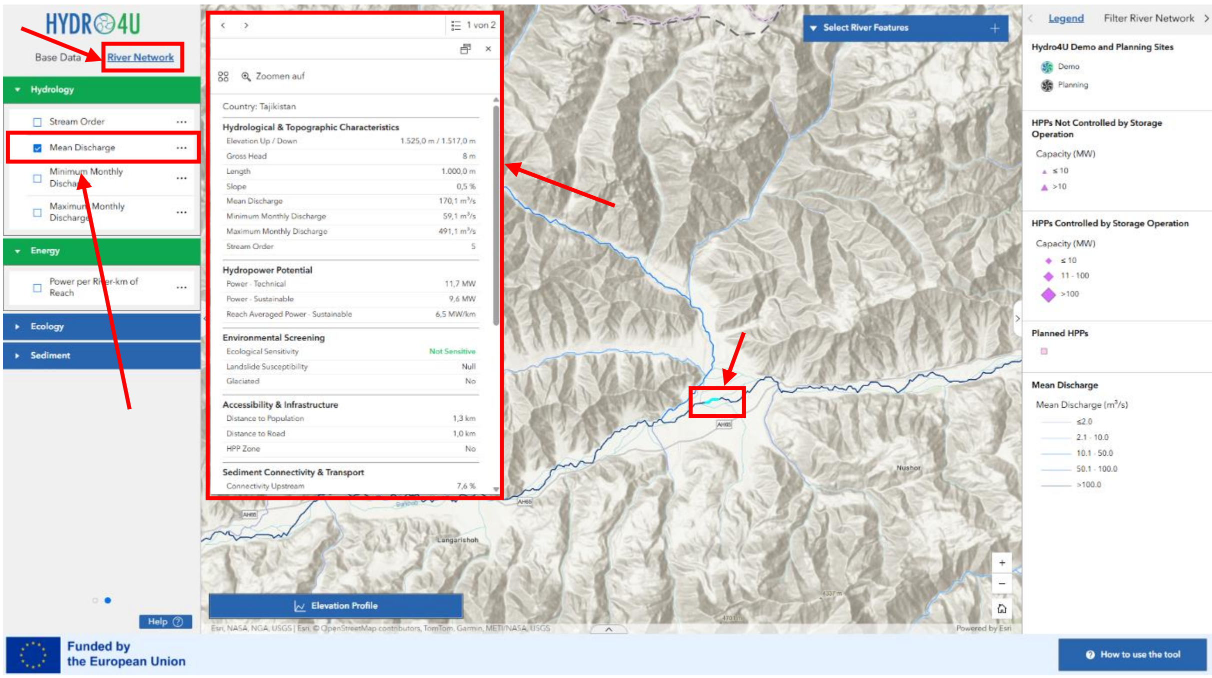
Note:

- *A river segment is defined as the **smallest spatial unit that can be selected**, whereas a **reach** is defined as **the sum of all segments between two confluences**. Unless stated otherwise, all information is provided at the **segment level**, not the reach level.*
- *Depending on the selected basemap, an **additional river network may be visible**. This basemap river network does not exactly match the calculated river network used in this tool and does not contain attribute data. **Only the calculated and selectable river network can be selected and queried.***
- *Within the **River Network** tab, the **Sediment layers** are displayed in three sub-groups: **Transport, Entrainment** and **Deposition Probabilities**. Each sub-group contains **5 layers**. To display these layers, the tick box for both the sub-group and the individual layer must be active.*



3.6 Legend

The legend is located in the **right sidebar** and displays all items that are currently shown on the map (both ***Base Data*** as well as information of the activated ***River Network***).

Note: *The Legend also provides information about the units of the datasets.*

3.7 Filter River Network

The ***Filter River Network*** function allows users to filter the river network based on predefined attributes. It is located in the **right sidebar** (see screenshot below), where users can switch between the **Legend** and the **Filter** view using the tabs at the top of the panel.

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The predefined filters are designed to help users identify suitable locations based on specific river characteristics.

To use the filters:

1. Select the ***Filter River Network*** tab on the top right of the screen.
2. Choose a numerical range or a predefined category for the selected attribute.
3. Activate the filter by toggling the **switch** to the top right of each filter.
4. The map will automatically zoom to the extent of the filtered river segments.

Multiple filters can be applied simultaneously. All active filters are combined, meaning only river segments that match **all selected criteria** will be displayed on the map.

River segments can be filtered according to the attributes listed below. These represent a selection of the attributes available in the river segment dataset. A description of each attribute can be found in Section 0.

- Country
- Average Slope of Reach* (%)
- Mean Discharge (m^3/s)
- Minimum Monthly Discharge (m^3/s)
- Maximum Monthly Discharge (m^3/s)
- Reach Averaged Power - Sustainable (MW/km)
- Distance to Population (km)
- Distance to Road (km)
- In Zone of Operation HPP
- Ecological Sensitivity
- Landslide Susceptibility
- Glaciated

**A reach is defined as the sum of river segments between two confluences.*

Note: *If fewer river segments than expected are displayed, check whether any filters are unintentionally active. A filter that is accidentally left on will reduce the results and may lead to incorrect conclusions. Make sure to deactivate all filters that are not needed. The option to deactivate all filters is available in the bottom-right corner.*

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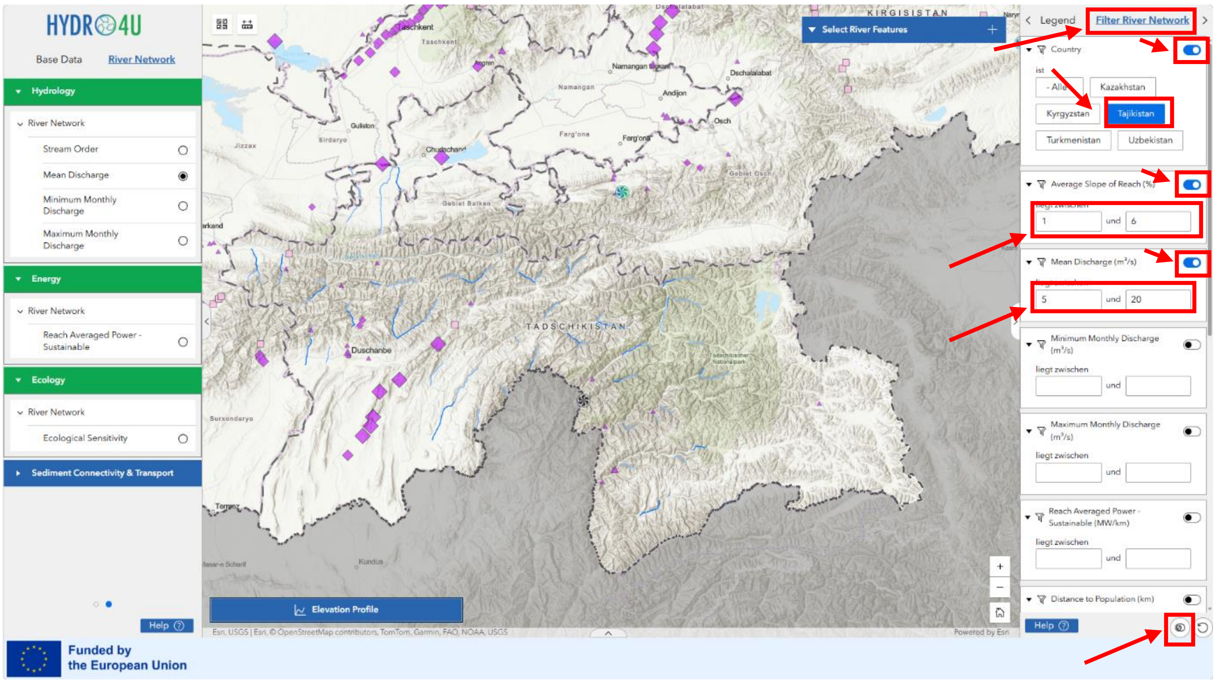
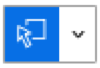
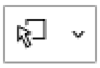


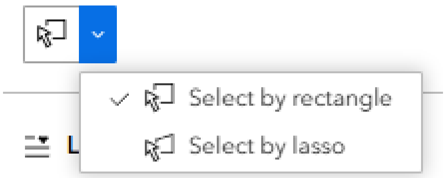
3.8 Select River Feature

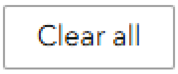
The Select River Feature allows the user to **select more than one river segments**.

To use the tool:

1. Expand the tool bar by clicking on the *Select River Feature* box.

2. Enable the selection by clicking on pointer icon. When it is blue it is active. Click on it again to deactivate the select feature tool.
   a.  Activated selection button.
   b.  Deactivated selection button.
3. The arrow to the right provides access to **Rectangle** or **Lasso** selection methods.



4. To select objects on the map, left-click several times to define the boundaries of the area in which the features should be selected.
5. To clear the selection, either use the  button or  button or click on an empty space on the map.

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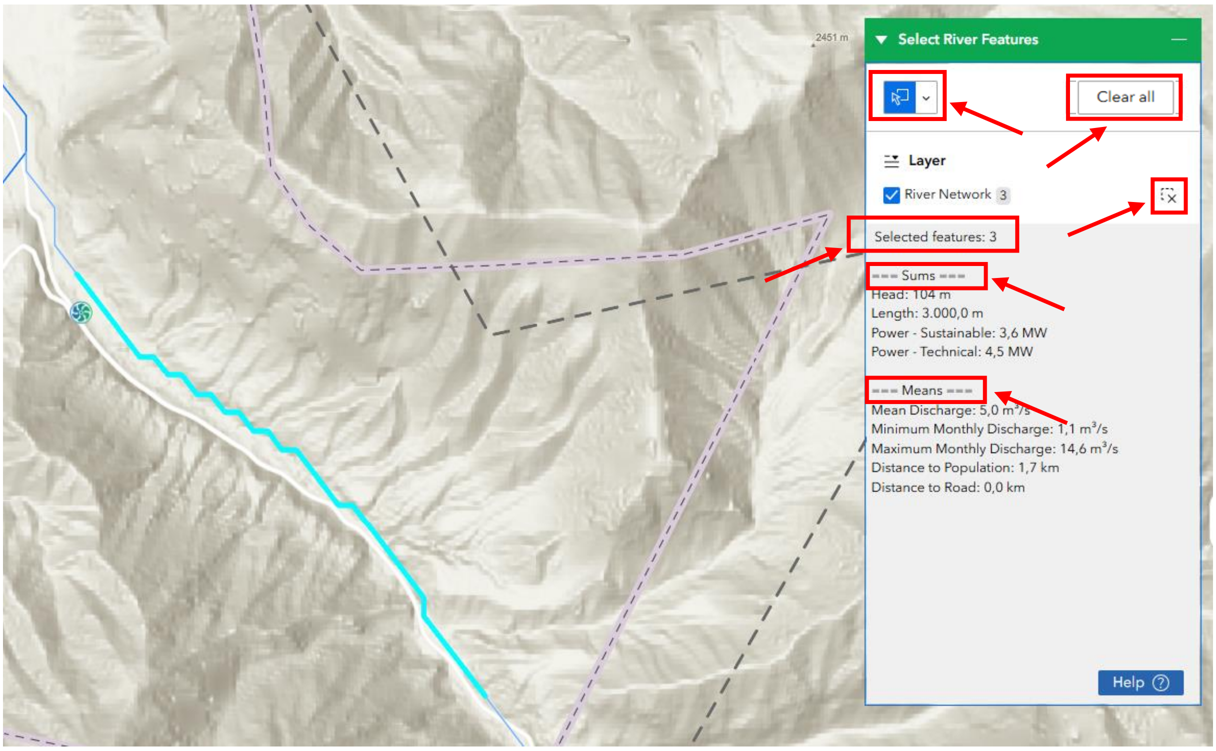
Once one or more river segments have been selected, the total number of segments, sum of attributes and means of attributes are automatically displayed in the box below.

The **Sums include**: *Head (m)*, *Length (m)*, *Power – Sustainable (MW)* and *Power – Technical (MW)* are displayed.

The **means include**: *Mean Discharge (m³/s)*, *Minimum Monthly Discharge (m³/s)*, Maximum Monthly Discharge (m³/s), *Distance to Population (km)* and *Distance to Road (km)*.



3.9 Elevation Profile

The ***Elevation Profile*** tool allows the user to generate an elevation profile graph of either a selected segment or segments of the ***River Network*** or from a line draw anywhere on the map.

Note: *The underlying DEM has a maximum resolution of 30 m and is therefore not suitable for detailed, site-specific analysis.*

1. Click the ***Elevation Profile*** button to open the elevation profile tool on the bottom of the screen. The size of the tool can be adjusted by clicking on the top of the box and adjusting to suit.

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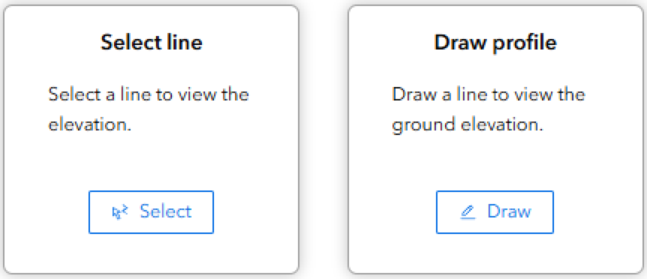
2. Choose one of the following options:



Option A: Select line

Click on a river segment to start and add the next upstream or downstream segment by clicking on the segment highlighted in yellow to extend the profile.

Note: *The Digital Elevation Model (DEM) used to generate the river network is not the same as the DEM used to generate on-the-fly elevation profiles. As a result, river segments are not automatically aligned with the thalweg (main flow path) of the DEM used for profile generation. We therefore recommend using the* ***Draw profile*** *option, for example, by digitizing the profile line based on the underlying aerial photo.*

Option B: Draw profile

Click freely in the map to draw a multi-point line, finish the line by double-clicking.

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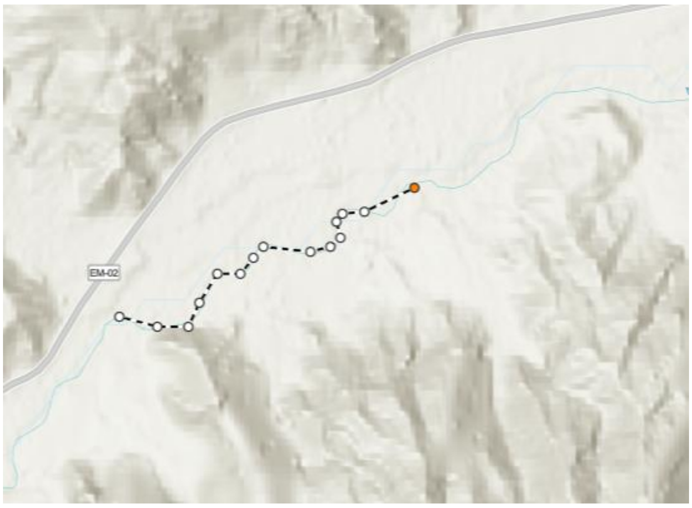
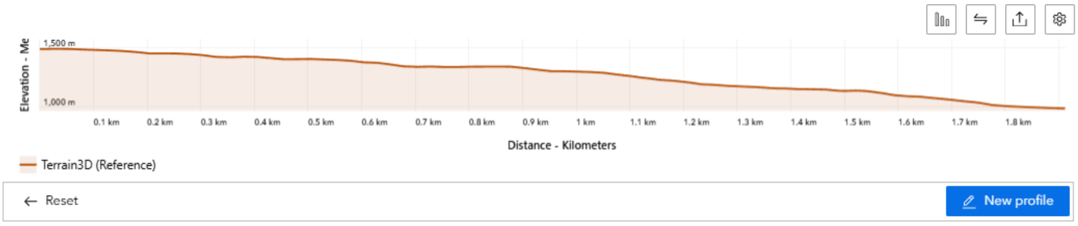
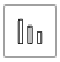


The elevation profile will be calculated automatically.



Note: *Click the* ← Reset *button in the lower left corner of the elevation profile window to clear the current profile and return to the option selection screen.*

3. In the top-right corner of the elevation profile window, the  button provides **basic statistics** for the generated profile.
4. Click either the Clear button or the New profile button in the lower right corner to **clear the current profile** and **start a new selection**.
5. Click the ***Elevation Profile*** button again to close the tool.

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